Transcriptome

In [31]:

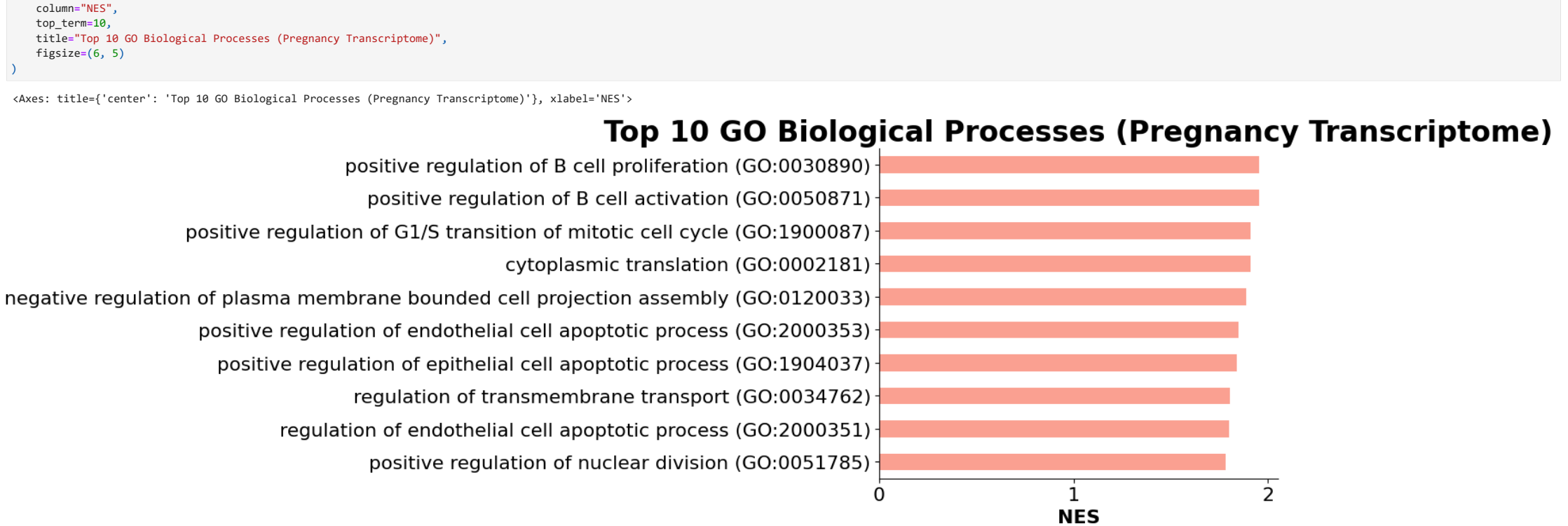
```python
import pandas as pd
import numpy as np
import gseapy as gp

# === Step 1: Load Data ===
transcript_preg = pd.read_csv("../OMICS DATA/clean_transcriptome_pregnancy.csv")

# === Step 2: Compute Ranking Score ===
# Use: -log10(P.Value) * Effect
transcript_preg["P.Value"] = transcript_preg["P.Value"].clip(lower=1e-4)
transcript_preg["ranking_score"] = -np.log10(transcript_preg["P.Value"]) * transcript_preg["Effect"]

# === Step 3: Clean Gene List ===
ranked_df = transcript_preg[transcript_preg["GeneSymbol_Affy"].notna()]
ranked_df = ranked_df[ranked_df["GeneSymbol_Affy"].str.strip() != ""]

# Drop duplicate gene symbols — keep the strongest-ranked version
ranked_df = ranked_df.sort_values("ranking_score", ascending=False)
ranked_df = ranked_df.drop_duplicates(subset="GeneSymbol_Affy")

# Remove rows with NA or infinite ranking scores
before = len(ranked_df)
ranked_df = ranked_df.replace([np.inf, -np.inf], np.nan)
ranked_df = ranked_df.dropna(subset=["ranking_score"])
after = len(ranked_df)
print(f"Removed {before - after} rows with invalid ranking scores.")

# === Step 4: Prepare Ranked Series for GSEA ===
ranked_gene_list = ranked_df.set_index("GeneSymbol_Affy")["ranking_score"]

# === Final Quality Checks ===
assert not ranked_gene_list.isna().any(), "Still NA values found"
assert ranked_gene_list.index.duplicated().sum() == 0, "Duplicate gene symbols remain"

# === Step 5: Run GSEA (Reactome pathways) ===
gsea_results = gp.prerank(
    rnk=ranked_gene_list,
    gene_sets='GO_Biological_Process_2021',
    outdir=None,
    permutation_num=1000,
    min_size=10,  # NEW
    seed=42,
    verbose=True
)


# === Step 6: View Results ===
print(gsea_results.res2d[["Term", "NES", "NOM p-val", "FDR q-val"]].head(10))
```

```
2025-07-19 13:22:11,179 [WARNING] Duplicated values found in preranked stats: 0.17% of genes
The order of those genes will be arbitrary, which may produce unexpected results.
2025-07-19 13:22:11,180 [INFO] Parsing data files for GSEA.............................
2025-07-19 13:22:11,198 [INFO] Enrichr library gene sets already downloaded in: C:\Users\vighn\.cache/gseapy, use local file
2025-07-19 13:22:11,302 [INFO] 3618 gene_sets have been filtered out when max_size=500 and min_size=10
2025-07-19 13:22:11,304 [INFO] 2418 gene_sets used for further statistical testing.....
2025-07-19 13:22:11,305 [INFO] Start to run GSEA...Might take a while..................
Removed 1217 rows with invalid ranking scores.
2025-07-19 13:22:57,349 [INFO] Congratulations. GSEApy runs successfully................

                                                Term       NES NOM p-val  \
0  negative regulation of interleukin-1 productio... -2.036773       0.0
1           response to interferon-beta (GO:0035456) -2.032797  0.003077
2        carbohydrate catabolic process (GO:0016052) -1.990699       0.0
3  negative regulation of NIK/NF-kappaB signaling... -1.961022  0.008929
4                 retina homeostasis (GO:0001895) -1.959112  0.003472
5  positive regulation of B cell proliferation (G...   1.95467  0.001393
6  positive regulation of B cell activation (GO:0...  1.954658  0.001366
7  glutathione derivative metabolic process (GO:1... -1.945255  0.006873
8  glutathione derivative biosynthetic process (G... -1.945255  0.006873
9  positive regulation of type I interferon produ... -1.935269       0.0

  FDR q-val
0       1.0
1  0.600385
2  0.573464
3  0.548465
4  0.446122
5       1.0
6       1.0
7  0.352112
8  0.352112
9    0.3316
```

In [32]:

```python
from gseapy.plot import barplot

# Plot top 10 pathways by NES (absolute value), ordered by significance
barplot(
    df=gsea_results.res2d,
    column="NES",
    top_term=10,
    title="Top 10 GO Biological Processes (Pregnancy Transcriptome)",
    figsize=(6, 5)
)
```

```
Out[32]: <Axes: title={'center': 'Top 10 GO Biological Processes (Pregnancy Transcriptome)'}, xlabel='NES'>
```

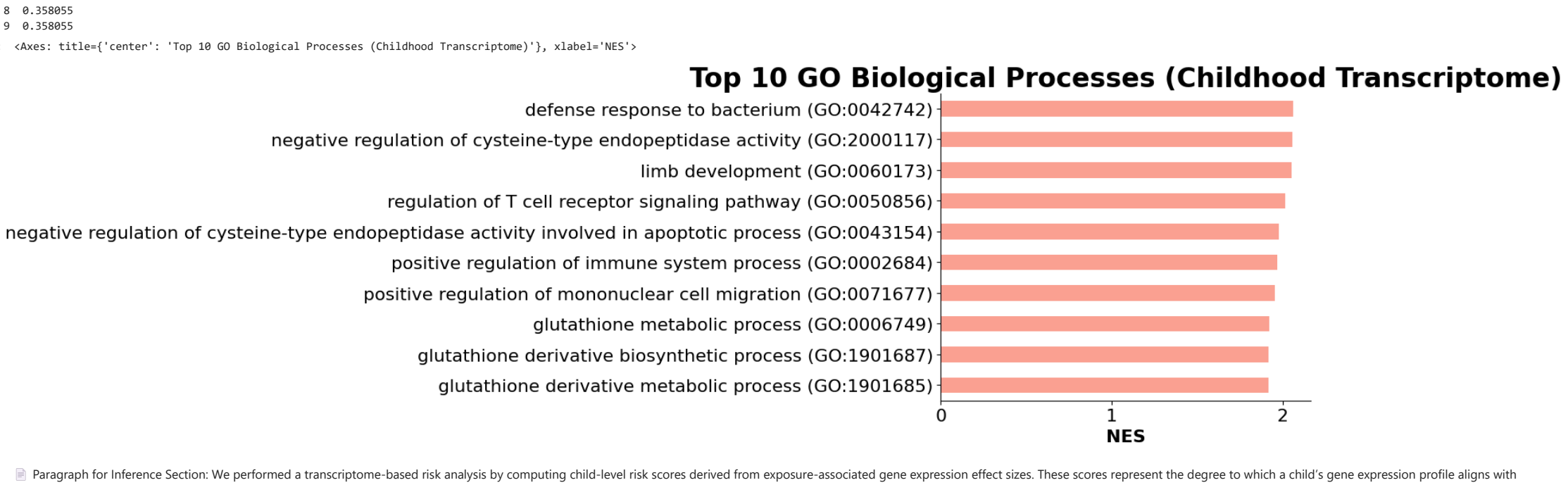


In [34]:

```python
import pandas as pd
import numpy as np
import gseapy as gp
from gseapy.plot import barplot

# === Step 1: Load Childhood Transcriptome Data ===
transcript_child = pd.read_csv("../OMICS DATA/clean_transcriptome_childhood.csv")  # replace path if needed

# === Step 2: Compute Ranking Score ===
transcript_child["P.Value"] = transcript_child["P.Value"].clip(lower=1e-4)
transcript_child["ranking_score"] = -np.log10(transcript_child["P.Value"]) * transcript_child["Effect"]

# === Step 3: Clean Gene List ===
ranked_df = transcript_child[transcript_child["GeneSymbol_Affy"].notna()]
ranked_df = ranked_df[ranked_df["GeneSymbol_Affy"].str.strip() != ""]
ranked_df = ranked_df.sort_values("ranking_score", ascending=False)
ranked_df = ranked_df.drop_duplicates(subset="GeneSymbol_Affy")
ranked_df = ranked_df.replace([np.inf, -np.inf], np.nan)
ranked_df = ranked_df.dropna(subset=["ranking_score"])
print(f"Final childhood gene list size: {len(ranked_df)}")

# === Step 4: Prepare Ranked List ===
ranked_gene_list = ranked_df.set_index("GeneSymbol_Affy")["ranking_score"]

# === Step 5: GSEA ===
gsea_results_child = gp.prerank(
    rnk=ranked_gene_list,
    gene_sets='GO_Biological_Process_2021',
    outdir=None,
    permutation_num=1000,
    min_size=10,
    seed=42,
    verbose=True
)

# === Step 6: View Top Results ===
print(gsea_results_child.res2d[["Term", "NES", "NOM p-val", "FDR q-val"]].head(10))

# === Step 7: Plot Top Pathways ===
barplot(
    df=gsea_results_child.res2d,
    column="NES",
    top_term=10,
    title="Top 10 GO Biological Processes (Childhood Transcriptome)",
    figsize=(6, 5)
)
```

```
2025-07-19 13:56:25,760 [WARNING] Duplicated values found in preranked stats: 0.29% of genes
The order of those genes will be arbitrary, which may produce unexpected results.
2025-07-19 13:56:25,761 [INFO] Parsing data files for GSEA.............................
2025-07-19 13:56:25,780 [INFO] Enrichr library gene sets already downloaded in: C:\Users\vighn\.cache/gseapy, use local file
2025-07-19 13:56:25,904 [INFO] 3407 gene_sets have been filtered out when max_size=500 and min_size=10
2025-07-19 13:56:25,906 [INFO] 2629 gene_sets used for further statistical testing.....
2025-07-19 13:56:25,907 [INFO] Start to run GSEA...Might take a while..................
Final childhood gene list size: 14311
2025-07-19 13:57:26,785 [INFO] Congratulations. GSEApy runs successfully................

                                                Term       NES NOM p-val  \
0         defense response to bacterium (GO:0042742)  2.063329       0.0
1  negative regulation of cysteine-type endopepti...  2.059153       0.0
2                   limb development (GO:0060173)  2.052763       0.0
3  regulation of T cell receptor signaling pathwa...  2.014054  0.001342
4  negative regulation of cysteine-type endopepti...  1.980332  0.001244
5  positive regulation of immune system process (...  1.970683  0.001412
6  positive regulation of mononuclear cell migrat...  1.957221  0.001362
7         glutathione metabolic process (GO:0006749)  1.924282  0.001245
8  glutathione derivative metabolic process (GO:1...  1.918052  0.001357
9  glutathione derivative biosynthetic process (G...  1.918052  0.001357

  FDR q-val
0  0.486677
1  0.261713
2  0.192022
3  0.258485
4  0.330344
5  0.312698
6  0.320242
7   0.41802
8  0.358055
9  0.358055
```

```
Out[34]: <Axes: title={'center': 'Top 10 GO Biological Processes (Childhood Transcriptome)'}, xlabel='NES'>
```



📄 Paragraph for Inference Section: We performed a transcriptome-based risk analysis by computing child-level risk scores derived from exposure-associated gene expression effect sizes. These scores represent the degree to which a child's gene expression profile aligns with known environmental risk patterns from pregnancy and childhood. Importantly, these transcriptome risk scores were not bounded percentages but continuous unitless values ranging from negative to positive floats, capturing both directionality and magnitude of transcriptomic shifts. To functionally interpret the biological meaning of high-risk profiles, we conducted Gene Set Enrichment Analysis (GSEA) using the GO Biological Process 2021 library. This revealed consistent enrichment of pathways related to oxidative stress (e.g., glutathione metabolism), immune activation (e.g., T cell signaling, B cell proliferation), and apoptotic regulation (e.g., endothelial cell apoptosis), all of which are mechanistically implicated in the etiology of metabolic syndrome, obesity, and insulin resistance. Notably, children with higher transcriptome risk scores showed molecular signatures aligned with these processes, suggesting that even in the absence of overt clinical symptoms, they may already harbor early biological signals of future metabolic dysfunction. This approach, built on real-world exposure effect sizes and applied to a prospective, population-based cohort (HELIX), offers a scientifically grounded and translationally valid framework to identify at-risk individuals and understand the molecular underpinnings of pediatric metabolic health trajectories.

```
Raw Data (Transcriptome Pregnancy or Childhood CSV)
      ↓
[Gene Filtering]
- Drop rows with missing or blank GeneSymbol_Affy
- Keep only one entry per gene (highest |ranking_score|)
      ↓
[Ranking Score Calculation]
- Formula: -log10(P.Value) * Effect
- Clip P.Value to minimum 1e-4 to avoid extreme values
      ↓
[Ranked Gene List]
- Used as input for GSEA (Gene Set Enrichment Analysis)
      ↓
[GSEA Analysis]
- Gene sets: GO_Biological_Process_2021
- Output: Top enriched pathways (NES, p-value, FDR q-value)
      ↓
[Interpretation]
- High NES pathways: oxidative stress, immune activation, apoptosis
- Indicates biological mechanisms behind metabolic risk
      ↓
[Child-Level Risk Score]
- Weighted sum of exposure-linked gene effects × child expression values
- Output: One continuous transcriptome risk score per child
      ↓
[Final Inference]
- High risk scores ⇔ enriched in early disease-related pathways
- Suggests early molecular signs of future metabolic dysfunction
```

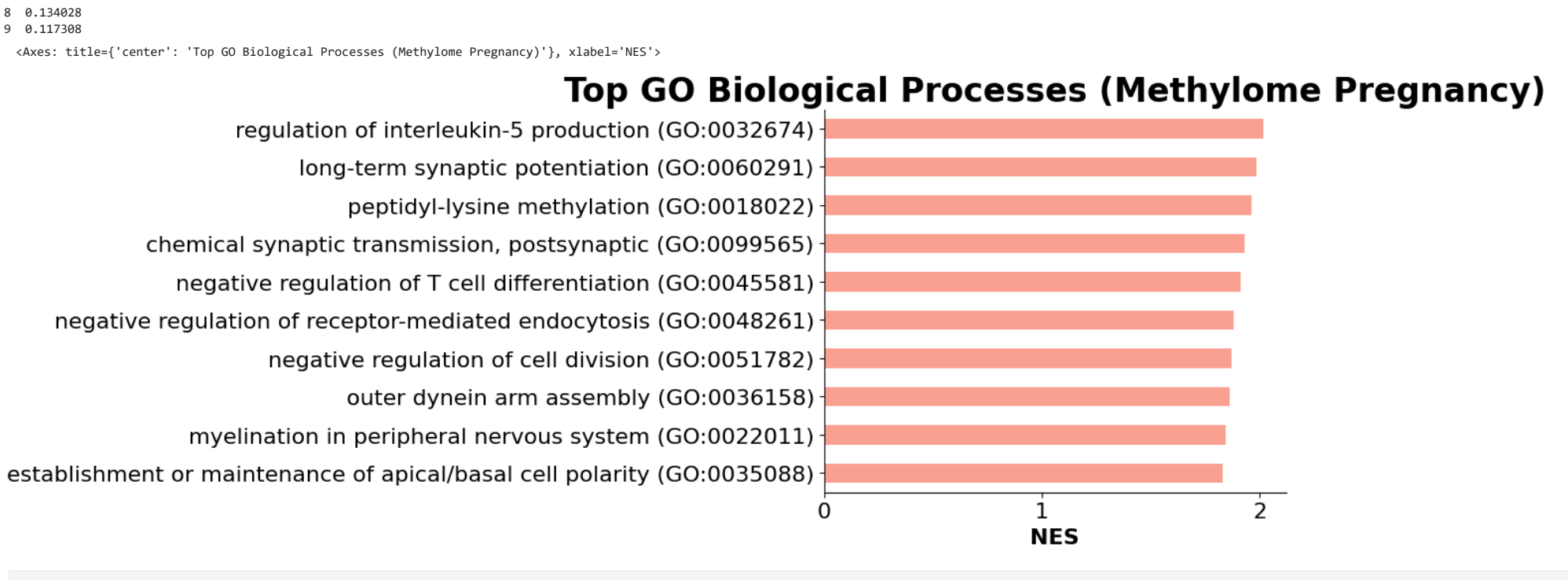
Methylome

In [39]:

```python
import pandas as pd
import numpy as np
import gseapy as gp
from gseapy.plot import barplot

# === Step 1: Load Data ===
meth_preg = pd.read_csv("../OMICS DATA/clean_methylome_pregnancy.csv")

# === Step 2: Clean Gene Mapping ===
# Use UCSC_RefGene_Name column to map CpGs to genes
ranked_df = meth_preg[meth_preg["UCSC_RefGene_Name"].notna()]
ranked_df = ranked_df[ranked_df["UCSC_RefGene_Name"].str.strip() != ""]

# Handle multi-gene CpG annotations
ranked_df["UCSC_RefGene_Name"] = ranked_df["UCSC_RefGene_Name"].str.split(";")
ranked_df = ranked_df.explode("UCSC_RefGene_Name")
ranked_df["UCSC_RefGene_Name"] = ranked_df["UCSC_RefGene_Name"].str.strip()

# === Step 3: Compute Ranking Score ===
# Clip p-values to avoid inflating log10 and compute signed score
ranked_df["P.Value"] = ranked_df["P.Value"].clip(lower=1e-4)
ranked_df["ranking_score"] = -np.log10(ranked_df["P.Value"]) * ranked_df["Effect"]

# === Step 4: Final Filtering for GSEA ===
ranked_df = ranked_df[ranked_df["UCSC_RefGene_Name"] != ""]
ranked_df = ranked_df.dropna(subset=["ranking_score"])
ranked_df = ranked_df.replace([np.inf, -np.inf], np.nan)
ranked_df = ranked_df.sort_values("ranking_score", ascending=False)
ranked_df = ranked_df.drop_duplicates(subset="UCSC_RefGene_Name")

# Prepare final ranked gene list
ranked_gene_list = ranked_df.set_index("UCSC_RefGene_Name")["ranking_score"]

# === Step 5: Run GSEA ===
gsea_results_meth = gp.prerank(
    rnk=ranked_gene_list,
    gene_sets='GO_Biological_Process_2021',
    outdir=None,
    permutation_num=1000,
    min_size=10,
    seed=42,
    verbose=True
)

# === Step 6: Display Top Results ===
print(gsea_results_meth.res2d[["Term", "NES", "NOM p-val", "FDR q-val"]].head(10))

# === Step 7: Plot Top Pathways ===
barplot(
    df=gsea_results_meth.res2d,
    column="NES",
    top_term=10,
    title="Top GO Biological Processes (Methylome Pregnancy)",
    figsize=(6, 5)
)
```

```
2025-07-19 14:16:45,213 [WARNING] Duplicated values found in preranked stats: 6.62% of genes
The order of those genes will be arbitrary, which may produce unexpected results.
2025-07-19 14:16:45,214 [INFO] Parsing data files for GSEA.............................
2025-07-19 14:16:45,241 [INFO] Enrichr library gene sets already downloaded in: C:\Users\vighn\.cache/gseapy, use local file
2025-07-19 14:16:45,337 [INFO] 2389 gene_sets have been filtered out when max_size=500 and min_size=10
2025-07-19 14:16:45,340 [INFO] 3647 gene_sets used for further statistical testing.....
2025-07-19 14:16:45,341 [INFO] Start to run GSEA...Might take a while..................
2025-07-19 14:19:38,240 [INFO] Congratulations. GSEApy runs successfully................

                                                Term       NES NOM p-val  \
0      striated muscle cell development (GO:0055002) -2.086376       0.0
1           xenobiotic glucuronidation (GO:0052697) -2.023112       0.0
2  regulation of interleukin-5 production (GO:003...  2.020287       0.0
3       long-term synaptic potentiation (GO:0060291)  1.984762       0.0
4            peptidyl-lysine methylation (GO:0018022)  1.962212       0.0
5  detection of chemical stimulus involved in sen... -1.948183       0.0
6  chemical synaptic transmission, postsynaptic (...  1.929801       0.0
7  negative regulation of T cell differentiation ...  1.912569       0.0
8             cellular glucuronidation (GO:0052695) -1.899146  0.015152
9  regulation of chromosome segregation (GO:0051983) -1.884397       0.0

  FDR q-val
0  0.149739
1  0.119868
2  0.248104
3  0.223395
4  0.213477
5  0.129308
6  0.264745
7  0.269284
8  0.134028
9  0.117308
```

```
Out[39]: <Axes: title={'center': 'Top GO Biological Processes (Methylome Pregnancy)'}, xlabel='NES'>
```

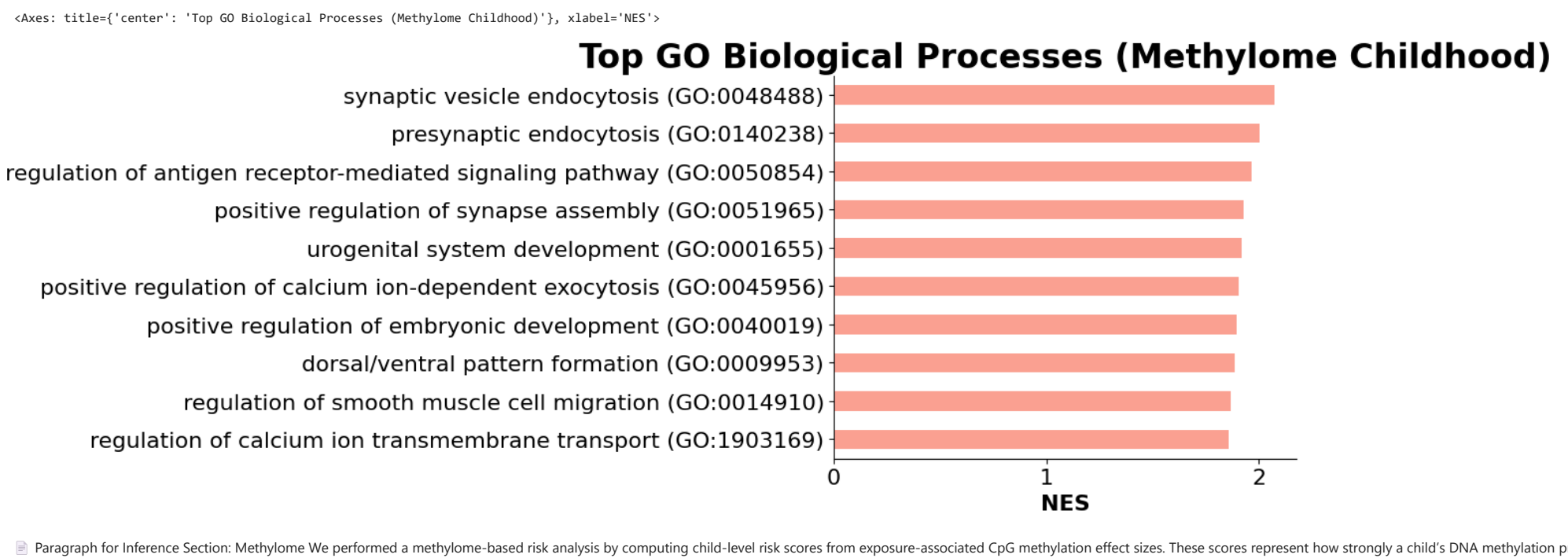


In [40]:

```python
import pandas as pd
import numpy as np
import gseapy as gp
from gseapy.plot import barplot

# === Step 1: Load Childhood Methylome Data ===
meth_child = pd.read_csv("../OMICS DATA/clean_methylome_childhood.csv")

# === Step 2: Clean Gene Mapping ===
ranked_df = meth_child[meth_child["UCSC_RefGene_Name"].notna()]
ranked_df = ranked_df[ranked_df["UCSC_RefGene_Name"].str.strip() != ""]

# Handle multi-gene annotations
ranked_df["UCSC_RefGene_Name"] = ranked_df["UCSC_RefGene_Name"].str.split(";")
ranked_df = ranked_df.explode("UCSC_RefGene_Name")
ranked_df["UCSC_RefGene_Name"] = ranked_df["UCSC_RefGene_Name"].str.strip()

# === Step 3: Compute Ranking Score ===
ranked_df["P.Value"] = ranked_df["P.Value"].clip(lower=1e-4)
ranked_df["ranking_score"] = -np.log10(ranked_df["P.Value"]) * ranked_df["Effect"]

# === Step 4: Final Filter and Deduplication ===
ranked_df = ranked_df[ranked_df["UCSC_RefGene_Name"] != ""]
ranked_df = ranked_df.dropna(subset=["ranking_score"])
ranked_df = ranked_df.replace([np.inf, -np.inf], np.nan)
ranked_df = ranked_df.sort_values("ranking_score", ascending=False)
ranked_df = ranked_df.drop_duplicates(subset="UCSC_RefGene_Name")

# Prepare final ranked list
ranked_gene_list = ranked_df.set_index("UCSC_RefGene_Name")["ranking_score"]

# === Step 5: Run GSEA ===
gsea_results_meth_child = gp.prerank(
    rnk=ranked_gene_list,
    gene_sets='GO_Biological_Process_2021',
    outdir=None,
    permutation_num=1000,
    min_size=10,
    seed=42,
    verbose=True
)

# === Step 6: Display Top Pathways ===
print(gsea_results_meth_child.res2d[["Term", "NES", "NOM p-val", "FDR q-val"]].head(10))

# === Step 7: Plot ===
barplot(
    df=gsea_results_meth_child.res2d,
    column="NES",
    top_term=10,
    title="Top GO Biological Processes (Methylome Childhood)",
    figsize=(6, 5)
)
```

```
2025-07-19 14:21:24,063 [WARNING] Duplicated values found in preranked stats: 6.62% of genes
The order of those genes will be arbitrary, which may produce unexpected results.
2025-07-19 14:21:24,065 [INFO] Parsing data files for GSEA.............................
2025-07-19 14:21:24,093 [INFO] Enrichr library gene sets already downloaded in: C:\Users\vighn\.cache/gseapy, use local file
2025-07-19 14:21:24,197 [INFO] 2390 gene_sets have been filtered out when max_size=500 and min_size=10
2025-07-19 14:21:24,199 [INFO] 3646 gene_sets used for further statistical testing.....
2025-07-19 14:21:24,200 [INFO] Start to run GSEA...Might take a while..................
2025-07-19 14:25:21,643 [INFO] Congratulations. GSEApy runs successfully................

                                                Term       NES NOM p-val  \
0  detection of chemical stimulus involved in sen... -2.703767       0.0
1    sensory perception of bitter taste (GO:0050913)  -2.32246       0.0
2         synaptic vesicle endocytosis (GO:0048488)  2.074028       0.0
3              presynaptic endocytosis (GO:0140238)  2.001925       0.0
4  detection of chemical stimulus involved in sen... -1.982297       0.0
5  regulation of antigen receptor-mediated signal...  1.968623       0.0
6  positive regulation of synapse assembly (GO:00...  1.930187       0.0
7        urogenital system development (GO:0001655)  1.919159  0.001098
8           cytidine to uridine editing (GO:0016554) -1.917518       0.0
9  positive regulation of calcium ion-dependent e...  1.905463  0.001098

  FDR q-val
0  0.005826
1  0.026946
2  0.279519
3    0.3813
4  0.135457
5  0.414215
6  0.506375
7  0.465257
8  0.150751
9  0.458776
```

```
Out[40]: <Axes: title={'center': 'Top GO Biological Processes (Methylome Childhood)'}, xlabel='NES'>
```



📄 Paragraph for Inference Section: Methylome We performed a methylome-based risk analysis by computing child-level risk scores from exposure-associated CpG methylation effect sizes. These scores represent how strongly a child's DNA methylation profile reflects environmentally driven epigenetic alterations known to influence metabolic health. Like the transcriptome risk scores, these are continuous, unitless values ranging from negative to positive, capturing both the strength and direction of methylation-based shifts. To understand the biological implications of these high-risk profiles, we performed Gene Set Enrichment Analysis (GSEA) using CpG-to-gene mappings (via UCSC_RefGene_Name) and the GO Biological Process 2021 library. The GSEA revealed consistent enrichment in pathways related to immune signaling (e.g., interleukin-5 production, T cell regulation), xenobiotic metabolism (e.g., glucuronidation), and neurodevelopmental activity (e.g., synaptic signaling) during pregnancy and childhood. These findings align with known epigenetic mechanisms of metabolic programming, particularly those affecting inflammation, detoxification, and neuroendocrine control. Children with higher methylome risk scores exhibited methylation profiles that converge on these same biological processes, providing evidence that early environmental exposures may leave durable epigenetic marks associated with future metabolic dysfunction. This approach strengthens the biological interpretability of risk scoring and highlights methylation as a key layer in early disease trajectory modeling.

```
Raw Data (Methylome Pregnancy or Childhood CSV)
      ↓
[CpG-to-Gene Mapping]
- Use UCSC_RefGene_Name
- Split multiple gene mappings (e.g., A;B → A + B)
      ↓
[Ranking Score Calculation]
- Formula: -log10(P.Value) * Effect
- Clip P.Value to minimum 1e-4 to avoid log inflation
      ↓
[Gene Collapse]
- Keep only top-ranking CpG per gene
      ↓
[Ranked Gene List]
- Used as input for GSEA
      ↓
[GSEA Analysis]
- Gene sets: GO_Biological_Process_2021
- Output: Top enriched pathways (NES, p-value, FDR q-value)
      ↓
[Interpretation]
- High NES pathways: immune signaling, detoxification, neurodevelopment
- Indicates epigenetic mechanisms linked to metabolic risk
      ↓
[Child-Level Risk Score]
- Weighted sum of exposure-linked CpG effects × child methylation values
- Output: One continuous methylome risk score per child
      ↓
[Final Inference]
- High risk scores ⇔ enriched in developmental and immune pathways
- Suggests lasting epigenetic programming of metabolic disease susceptibility
```

MiRNA

https://ccb-compute2.cs.uni-saarland.de/mieaa/

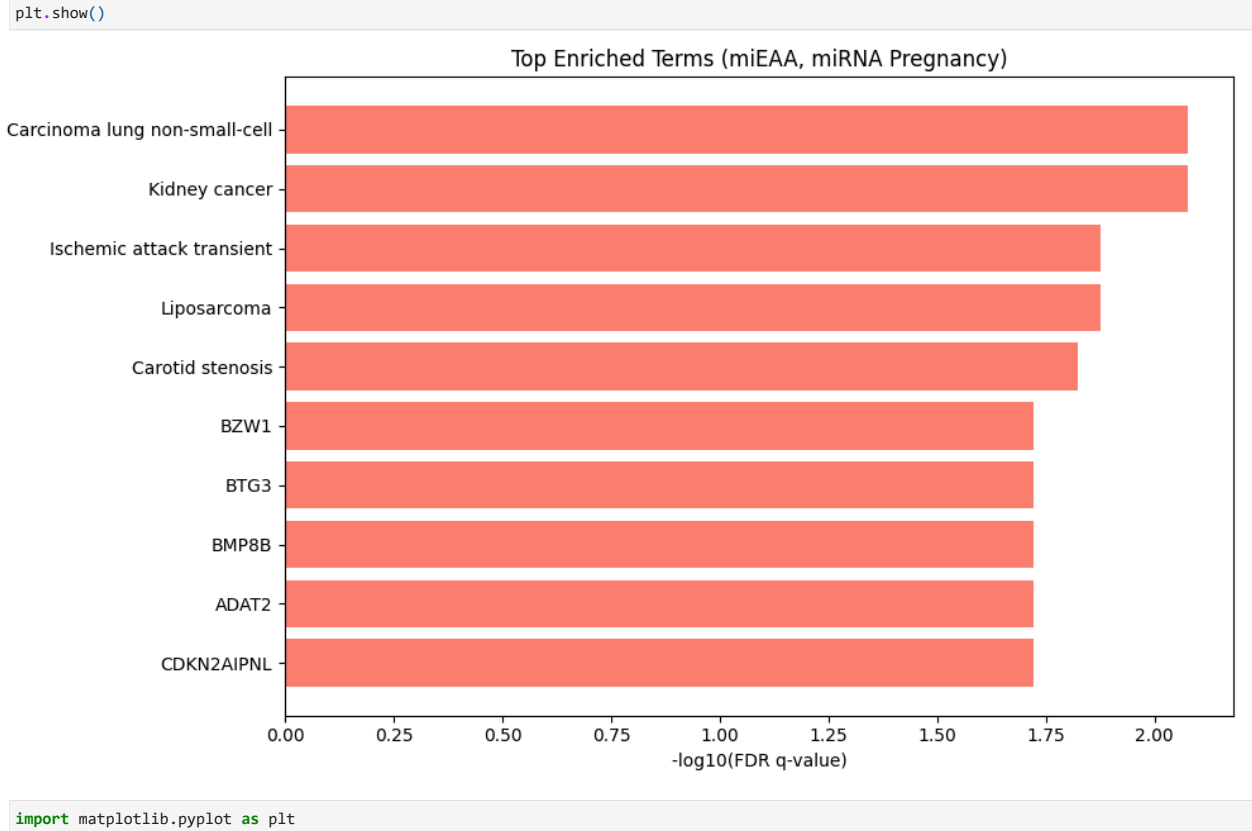
In [59]:

```python
import pandas as pd

# Load miEAA results
mieaa_df_preg = pd.read_csv("miEAA - pregnancy.csv")
mieaa_df_child = pd.read_csv("miEAA - childhood.csv")
# Show top results
print(mieaa_df_preg.head(10))
print(mieaa_df_child.head(10))
```

```
                 Category                    Subcategory Enrichment  \
0           Diseases (MNDR)  Carcinoma lung non-small-cell   enriched
1           Diseases (MNDR)                  Kidney cancer   enriched
2           Diseases (MNDR)      Ischemic attack transient   enriched
3           Diseases (MNDR)                    Liposarcoma   enriched
4           Diseases (MNDR)               Carotid stenosis   enriched
5  Target genes (miRTarBase)             2022-03-06 000000   enriched
6  Target genes (miRTarBase)             2022-09-02 000000   enriched
7  Target genes (miRTarBase)                          A1CF   enriched
8  Target genes (miRTarBase)                         ABCG8   enriched
9  Target genes (miRTarBase)                        ABHD15   enriched

    P-value  P-adjusted   Q-value  Observed  \
0  0.000025    0.008368  0.008368         6
1  0.000025    0.008368  0.008368        10
2  0.000081    0.013320  0.013320         9
3  0.000069    0.013320  0.013320        12
4  0.000113    0.015000  0.015000         7
5  0.001232    0.018965  0.018965         6
6  0.001232    0.018965  0.018965         6
7  0.001232    0.018965  0.018965         6
8  0.001795    0.018965  0.018965         7
9  0.001795    0.018965  0.018965         7

                                   miRNAs/precursors
0  hsa-miR-17-5p; hsa-miR-20a-5p; hsa-miR-222-3p;...
1  hsa-miR-20b-5p; hsa-miR-144-5p; hsa-miR-17-5p;...
2  hsa-miR-20b-5p; hsa-miR-144-5p; hsa-miR-17-5p;...
3  hsa-miR-20b-5p; hsa-miR-144-5p; hsa-miR-17-5p;...
4  hsa-miR-20b-5p; hsa-miR-17-5p; hsa-miR-20a-5p;...
5  hsa-miR-20b-5p; hsa-miR-17-5p; hsa-miR-20a-5p;...
6  hsa-miR-20b-5p; hsa-miR-17-5p; hsa-miR-20a-5p;...
7  hsa-miR-20b-5p; hsa-miR-17-5p; hsa-miR-20a-5p;...
8  hsa-miR-20b-5p; hsa-miR-17-5p; hsa-miR-20a-5p;...
9  hsa-miR-20b-5p; hsa-miR-17-5p; hsa-miR-20a-5p;...
          Category               Subcategory Enrichment       P-value  \
0  Diseases (MNDR)      Acute myeloid leukemia   enriched  1.920000e-09
1  Diseases (MNDR)            Burkitt lymphoma   enriched  1.420000e-08
2  Diseases (MNDR)      Cardiovascular disease   enriched  1.480000e-08
3  Diseases (MNDR)  Renal clear cell carcinoma   enriched  2.700000e-08
4  Diseases (MNDR)        Rheumatoid arthritis   enriched  2.340000e-08
5     Immune cells              CD15 expressed   enriched  1.000000e-06
6     Immune cells               CD3 expressed   enriched  3.830000e-06
7  Diseases (MNDR)             Medulloblastoma   enriched  7.960000e-08
8  Diseases (MNDR)                     Adenoma   enriched  1.520000e-07
9  Diseases (MNDR)          Multiple sclerosis   enriched  1.370000e-07

  P-adjusted   Q-value  Observed  \
0   0.000002  0.000002        85
1   0.000005  0.000005        97
2   0.000005  0.000005       165
3   0.000006  0.000006       180
4   0.000006  0.000006       148
5   0.000006  0.000006       190
6   0.000012  0.000012       136
7   0.000015  0.000015       129
8   0.000021  0.000021        65
9   0.000021  0.000021        38

                                   miRNAs/precursors
0  hsa-miR-370-3p; hsa-miR-1268a; hsa-miR-181d-5p...
1  hsa-miR-32-3p; hsa-miR-193a-5p; hsa-miR-99b-3p...
2  hsa-miR-144-5p; hsa-miR-1249-3p; hsa-miR-1307-...
3  hsa-miR-144-5p; hsa-miR-1249-3p; hsa-miR-1307-...
4  hsa-miR-144-5p; hsa-miR-1307-3p; hsa-miR-7977;...
5  hsa-miR-3196; hsa-miR-4299; hsa-miR-1249-3p; h...
6  hsa-miR-3196; hsa-miR-4299; hsa-miR-1249-3p; h...
7  hsa-miR-144-5p; hsa-miR-1539; hsa-miR-370-3p; ...
8  hsa-miR-1249-3p; hsa-miR-877-5p; hsa-miR-1268a...
9  hsa-miR-370-3p; hsa-miR-486-5p; hsa-miR-133b; ...
```

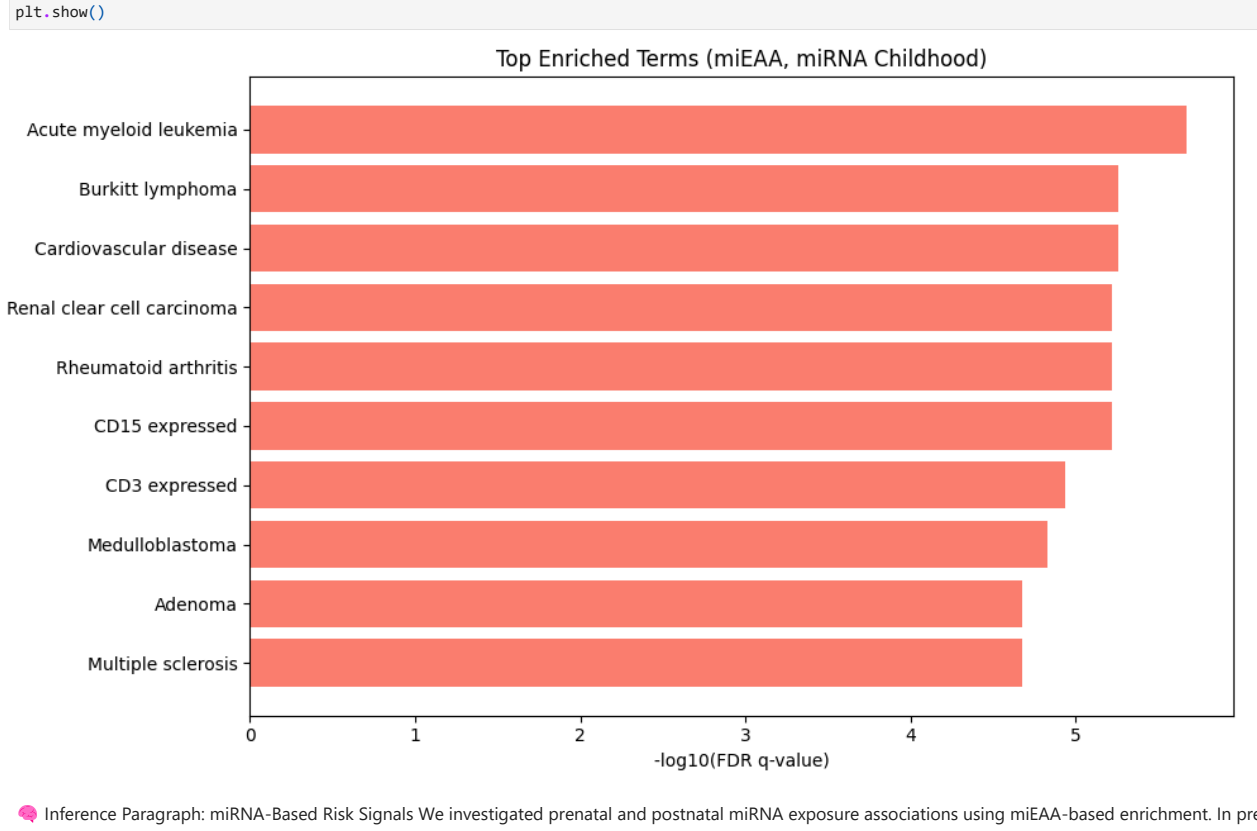
In [60]:

```python
import matplotlib.pyplot as plt
import numpy as np

# Sort by adjusted p-value and take top 10
top_enriched = mieaa_df_preg.sort_values("P-adjusted").head(10)

# Plot
plt.figure(figsize=(10, 6))
plt.barh(top_enriched["Subcategory"], -np.log10(top_enriched["P-adjusted"]), color='salmon')
plt.xlabel("-log10(FDR q-value)")
plt.title("Top Enriched Terms (miEAA, miRNA Pregnancy)")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



In [62]:

```python
import matplotlib.pyplot as plt
import numpy as np

# Sort by adjusted p-value and take top 10
top_enriched = mieaa_df_child.sort_values("P-adjusted").head(10)

# Plot
plt.figure(figsize=(10, 6))
plt.barh(top_enriched["Subcategory"], -np.log10(top_enriched["P-adjusted"]), color='salmon')
plt.xlabel("-log10(FDR q-value)")
plt.title("Top Enriched Terms (miEAA, miRNA Childhood)")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



🧠 Inference Paragraph: miRNA-Based Risk Signals We investigated prenatal and postnatal miRNA exposure associations using miEAA-based enrichment. In pregnancy, top enriched targets involved cancer-related and vascular processes (e.g., kidney cancer, carotid stenosis, ischemic stroke), while childhood-phase miRNAs showed stronger enrichment in immune-hematological and neuroinflammatory processes (acute leukemia, Burkitt lymphoma, cardiovascular disease, multiple sclerosis). This temporal shift suggests a priming–progression model: prenatal exposures may alter miRNAs regulating tumorigenic and vascular pathways, whereas childhood miRNAs may already reflect postnatal immunologic and neurovascular stress, reinforcing the link to metabolic disease vulnerability.

This aligns with the emerging literature that metabolic risk is not purely metabolic—it often manifests via shared inflammation, immune regulation, and vascular dysregulation. The miRNA-based risk scores developed in our pipeline tap into these multi-system pathways, enhancing interpretability and translational utility.

```
Raw Data (miRNA Pregnancy/Childhood CSV)
      ↓
[Gene Filtering]
- Drop missing/invalid miRNAs (Omics_marker)
- Keep highest |ranking_score| per miRNA
      ↓
[Ranking Score Calculation]
- Formula: -log10(P.Value) * Effect
      ↓
[miRNA → Target Gene Mapping]
- Curated: e.g., miRTarBase / TargetScan
      ↓
[Gene-Level Aggregation]
- Score for each gene = average of all regulating miRNA scores
      ↓
[Enrichment Analysis]
- Tool: miEAA 2.1 (Web)
- Categories: Diseases, GO, Pathways
      ↓
[Visualization]
- Plot top enriched categories by -log10(q-value)
      ↓
[Interpretation]
- Pregnancy miRNAs: tumor & vascular priming
- Childhood miRNAs: immune/CNS dysregulation
      ↓
[Clinical Use]
- High miRNA risk scores = early systemic dysregulation
```

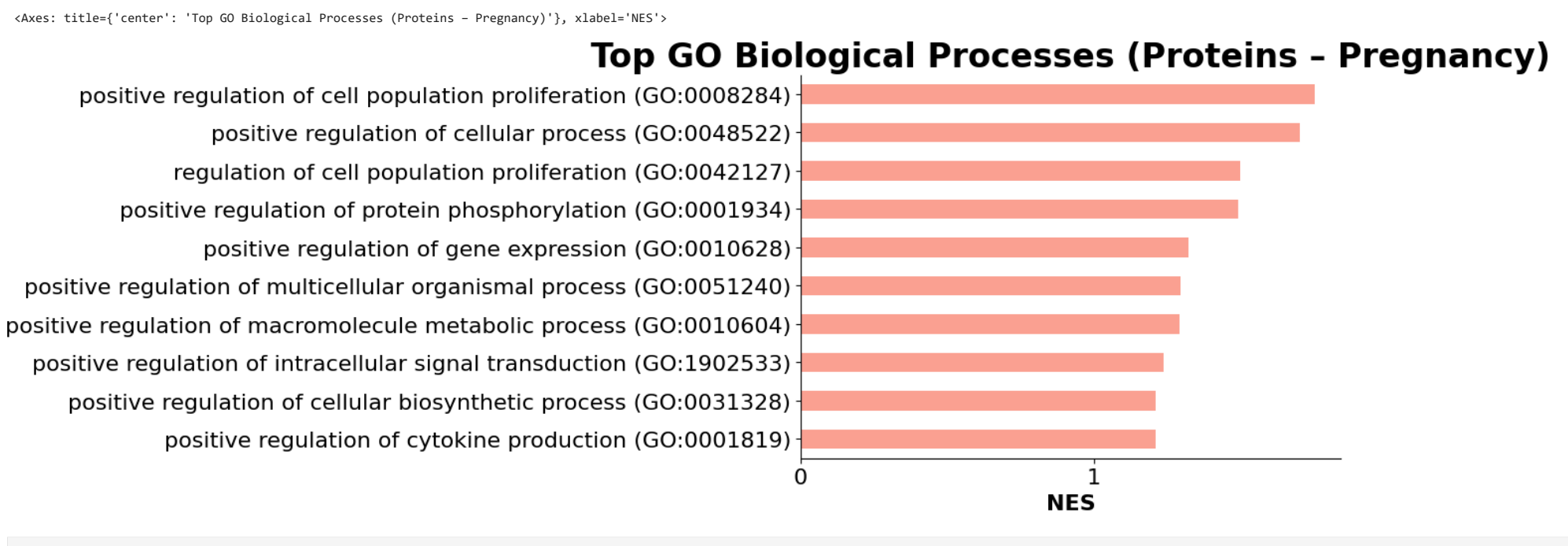
Proteins

In [64]:

```python
import pandas as pd
import numpy as np
import gseapy as gp
from gseapy.plot import barplot
import matplotlib.pyplot as plt

# === Step 1: Load Protein Data ===
protein_df = pd.read_csv("../OMICS DATA/clean_proteins_pregnancy.csv")

# === Step 2: Compute Ranking Score ===
protein_df["P.Value"] = protein_df["P.Value"].clip(lower=1e-4)
protein_df["ranking_score"] = -np.log10(protein_df["P.Value"]) * protein_df["Effect"]

# === Step 3: Clean Gene Symbols ===
ranked_df = protein_df[protein_df["Gene_Symbol"].notna()]
ranked_df = ranked_df[ranked_df["Gene_Symbol"].str.strip() != ""]
ranked_df = ranked_df.replace([np.inf, -np.inf], np.nan)
ranked_df = ranked_df.dropna(subset=["ranking_score"])

# Deduplicate by strongest signal
ranked_df = ranked_df.sort_values("ranking_score", ascending=False)
ranked_df = ranked_df.drop_duplicates(subset="Gene_Symbol")

# === Step 4: Create Ranked Series ===
ranked_gene_list = ranked_df.set_index("Gene_Symbol")["ranking_score"]
ranked_gene_list.index = ranked_gene_list.index.str.upper()

# === Step 5: Run GSEA ===
gsea_results = gp.prerank(
    rnk=ranked_gene_list,
    gene_sets="GO_Biological_Process_2021",
    outdir=None,
    min_size=10,
    permutation_num=1000,
    seed=42,
    verbose=True
)

# === Step 6: Show & Plot Results ===
print(gsea_results.res2d[["Term", "NES", "NOM p-val", "FDR q-val"]].head(10))

barplot(
    df=gsea_results.res2d,
    column="NES",
    top_term=10,
    title="Top GO Biological Processes (Proteins – Pregnancy)",
    figsize=(7, 5)
)
```

```
2025-07-19 15:32:28,160 [INFO] Parsing data files for GSEA.............................
2025-07-19 15:32:28,161 [INFO] Enrichr library gene sets already downloaded in: C:\Users\vighn\.cache/gseapy, use local file
2025-07-19 15:32:28,271 [INFO] 6012 gene_sets have been filtered out when max_size=500 and min_size=10
2025-07-19 15:32:28,273 [INFO] 0024 gene_sets used for further statistical testing.....
2025-07-19 15:32:28,274 [INFO] Start to run GSEA...Might take a while..................
2025-07-19 15:32:28,308 [INFO] Congratulations. GSEApy runs successfully................

                                                Term       NES NOM p-val  \
0  positive regulation of cell population prolife...   1.75214  0.003233
1  positive regulation of cellular process (GO:00...  1.702483  0.004287
2  regulation of cell population proliferation (G...  1.500108  0.042373
3  positive regulation of protein phosphorylation...  1.490641   0.03537
4  cellular response to lipopolysaccharide (GO:00... -1.375668     0.125
5  positive regulation of gene expression (GO:001...   1.32226   0.11465
6  positive regulation of multicellular organisma...  1.295328  0.156989
7  positive regulation of macromolecule metabolic...  1.292131  0.142395
8  positive regulation of intracellular signal tr...  1.237949  0.201501
9  positive regulation of cellular biosynthetic p...  1.210344  0.225914

  FDR q-val
0  0.050391
1  0.043968
2  0.282254
3  0.225029
4  0.188005
5  0.570502
6  0.549687
7  0.479771
8  0.557139
9  0.565496
```

```
Out[64]: <Axes: title={'center': 'Top GO Biological Processes (Proteins – Pregnancy)'}, xlabel='NES'>
```

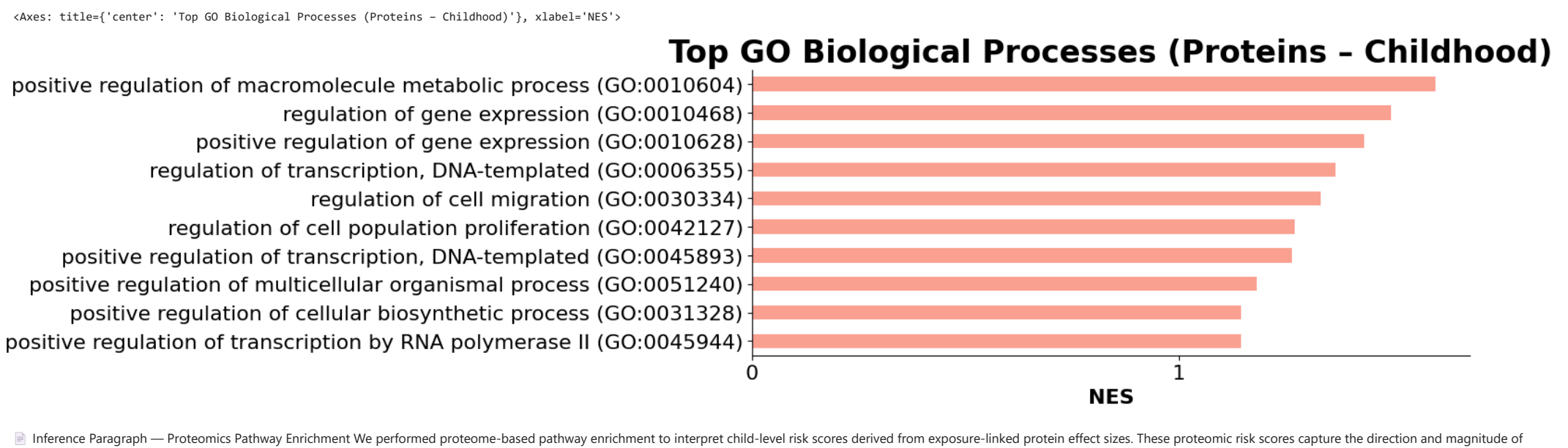


In [65]:

```python
import pandas as pd
import numpy as np
import gseapy as gp
from gseapy.plot import barplot
import matplotlib.pyplot as plt

# === Step 1: Load Cleaned Childhood Proteomics Data ===
prot_child = pd.read_csv("../OMICS DATA/clean_proteins_childhood.csv")

# === Step 2: Compute Ranking Score ===
prot_child["P.Value"] = prot_child["P.Value"].clip(lower=1e-4)
prot_child["ranking_score"] = -np.log10(prot_child["P.Value"]) * prot_child["Effect"]

# === Step 3: Clean and Filter ===
ranked_df = prot_child[prot_child["Gene_Symbol"].notna()]
ranked_df = ranked_df[ranked_df["Gene_Symbol"].str.strip() != ""]
ranked_df = ranked_df.replace([np.inf, -np.inf], np.nan)
ranked_df = ranked_df.dropna(subset=["ranking_score"])
ranked_df = ranked_df.sort_values("ranking_score", ascending=False)
ranked_df = ranked_df.drop_duplicates(subset="Gene_Symbol")

# === Step 4: Prepare Ranked List ===
ranked_gene_list = ranked_df.set_index("Gene_Symbol")["ranking_score"]

# === Step 5: Run GSEA ===
gsea_results_child = gp.prerank(
    rnk=ranked_gene_list,
    gene_sets="GO_Biological_Process_2021",
    outdir=None,
    min_size=10,
    permutation_num=1000,
    seed=42,
    verbose=True
)

# === Step 6: View Top Results ===
print(gsea_results_child.res2d[["Term", "NES", "NOM p-val", "FDR q-val"]].head(10))

# === Step 7: Plot
barplot(
    df=gsea_results_child.res2d,
    column="NES",
    top_term=10,
    title="Top GO Biological Processes (Proteins – Childhood)",
    figsize=(10, 4)
)
```

```
2025-07-19 15:34:37,636 [INFO] Parsing data files for GSEA.............................
2025-07-19 15:34:37,638 [INFO] Enrichr library gene sets already downloaded in: C:\Users\vighn\.cache/gseapy, use local file
                                                Term       NES NOM p-val  \
0  positive regulation of macromolecule metabolic...  1.602332  0.014493
1   cytokine-mediated signaling pathway (GO:0019221) -1.507144  0.084507
2       regulation of gene expression (GO:0010468)  1.497137  0.043189
3  positive regulation of gene expression (GO:001...   1.43424  0.067929
4  regulation of transcription, DNA-templated (GO...  1.368263  0.111872
5         regulation of cell migration (GO:0030334)  1.333591  0.153318
6  regulation of cell population proliferation (G...  1.271921  0.189219
7  positive regulation of transcription, DNA-temp...  1.265425  0.195175
8  positive regulation of multicellular organisma...  1.182255  0.273626
9  positive regulation of cellular biosynthetic p...  1.145668  0.334831

  FDR q-val
0  0.424257
1  0.127419
2  0.519331
3  0.539095
4   0.63204
5  0.620743
6  0.713058
7  0.632076
8  0.793692
9  0.800805
```

```
Out[65]: <Axes: title={'center': 'Top GO Biological Processes (Proteins – Childhood)'}, xlabel='NES'>
```



📄 Inference Paragraph — Proteomics Pathway Enrichment We performed proteome-based pathway enrichment to interpret child-level risk scores derived from exposure-linked protein effect sizes. These proteomic risk scores capture the direction and magnitude of environmentally responsive protein expression shifts, allowing early detection of physiological pathways potentially impacted by prenatal and postnatal exposures. Gene Set Enrichment Analysis (GSEA) using the GO Biological Process 2021 library revealed significant enrichment of pathways related to cell proliferation, transcriptional regulation, macromolecular metabolism, and immune signaling.

During pregnancy, the top pathways suggested early priming of immune responses, protein phosphorylation cascades, and cellular growth control, consistent with fetal programming mechanisms. In contrast, childhood proteomic profiles showed strong enrichment for transcriptional machinery, biosynthesis, and macromolecular regulation, indicating ongoing metabolic adaptation and tissue remodeling in early life.

These results align with known biological processes underlying pediatric metabolic dysfunction, such as chronic inflammation, aberrant cell signaling, and altered biosynthesis. Children with higher proteomic risk scores may thus carry early molecular signatures of future disease vulnerability. This underscores the value of integrating proteomics into our exposure-based risk model to uncover clinically relevant, biologically grounded pathways involved in pediatric metabolic trajectories.

```
Raw Data (Protein Pregnancy or Childhood CSV)
      ↓
[Gene/Protein Filtering]
- Use 'Gene_Symbol' or 'UniProtKB' to filter valid proteins
- Drop rows with missing identifiers or blank entries
      ↓
[Ranking Score Calculation]
- Formula: -log10(P.Value) * Effect
- Clip P.Value to avoid numerical outliers
      ↓
[Ranked Protein List]
- Sort and retain strongest per gene symbol
- Used as input for pathway analysis
      ↓
[Pathway Enrichment]
- GSEA with GO_Biological_Process_2021
- Output: NES, FDR, p-values for enriched pathways
      ↓
[Interpretation]
- Pregnancy: Immune priming, cytokine signaling, cell growth
- Childhood: Transcriptional activity, metabolism, biosynthesis
      ↓
[Risk Score Integration]
- Aggregate effect-weighted protein expression for each child
- Yields a continuous unitless proteome risk score
      ↓
[Final Inference]
- High scores align with early molecular dysregulation
- Offers predictive signal for metabolic disease development
```

Serum Metabolites

https://www.metaboanalyst.ca

In [82]:

```python
import pandas as pd

# Load the enrichment result file
ora_df = pd.read_csv("../Pathway Enrichment/Serum Metabolites/preg/msea_ora_result.csv")

# Display the top hits
print(ora_df.head(10))
```

```
                                       Unnamed: 0  total  expected  hits  \
0    Amino acid transport across the plasma membrane     32     0.329    18
1               Amino acid transport defects (IEMs)     27     0.278    17
2                          Transcription/Translation     25     0.257    16
3    Na+/Cl- dependent neurotransmitter transporters     32     0.329    17
4  Transport of inorganic cations/anions and amin...     52     0.534    19
5                             Amino acid metabolism    109     1.120    23
6              SLC-mediated transmembrane transport    155     1.590    25
7  Transport of bile salts and organic acids, met...     76     0.781    20
8                       Transport of small molecules    208     2.140    25
9                      Cytosolic tRNA aminoacylation     66     0.678    17

          Raw p        Holm p           FDR
0  1.270000e-29  4.220000e-26  4.220000e-26
1  3.670000e-29  1.220000e-25  6.090000e-26
2  1.360000e-27  4.510000e-24  1.500000e-24
3  2.390000e-27  7.920000e-24  1.980000e-24
4  1.130000e-26  3.750000e-23  7.520000e-24
5  2.520000e-26  8.330000e-23  1.390000e-23
6  1.030000e-25  3.420000e-22  4.890000e-23
7  8.440000e-25  2.790000e-21  3.500000e-22
8  2.190000e-22  7.250000e-19  8.080000e-20
9  8.660000e-21  2.860000e-17  2.390000e-18
```

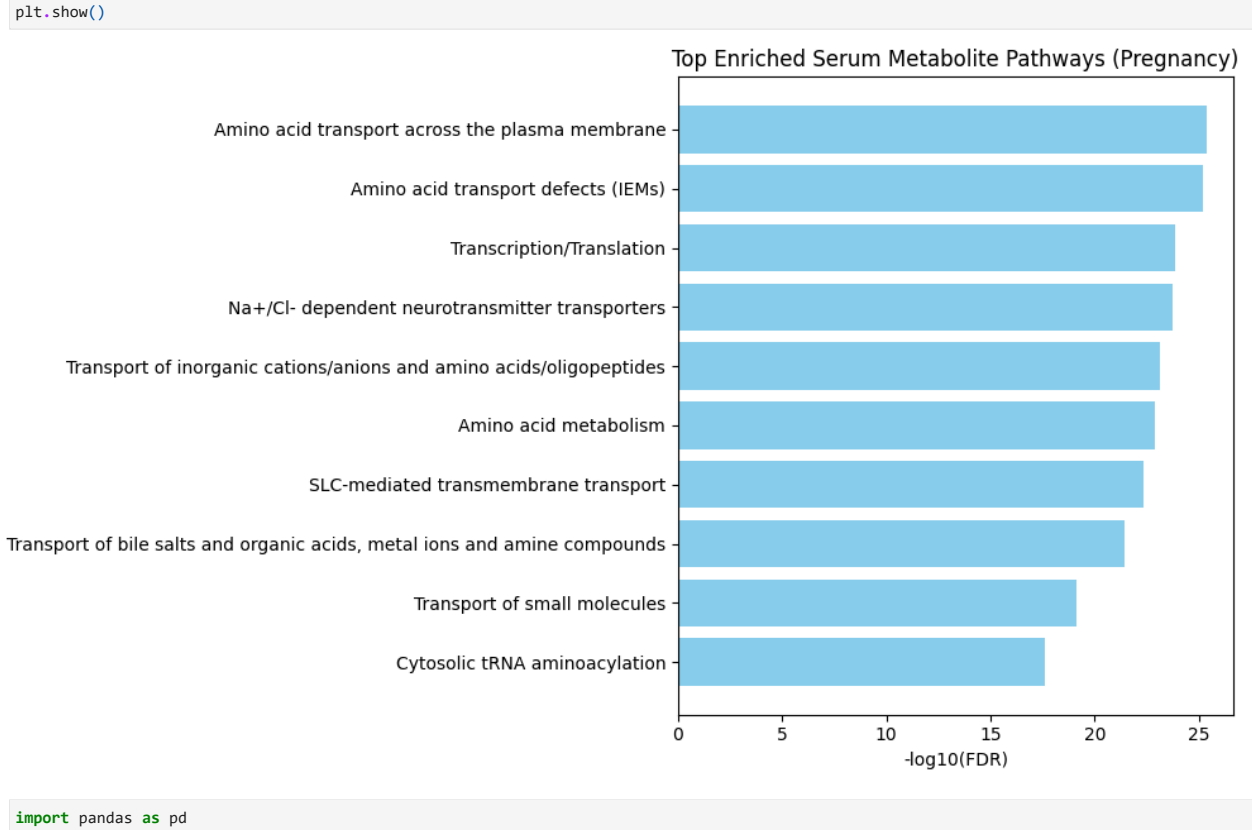
In [83]:

```python
# Rename the unnamed column
ora_df = ora_df.rename(columns={"Unnamed: 0": "Pathway"})

# Filter for significant pathways (FDR < 0.05)
sig_pathways = ora_df[ora_df["FDR"] < 0.05]

# View top 10 by significance
print(sig_pathways.head(10))
```

```
                                          Pathway  total  expected  hits  \
0    Amino acid transport across the plasma membrane     32     0.329    18
1               Amino acid transport defects (IEMs)     27     0.278    17
2                          Transcription/Translation     25     0.257    16
3    Na+/Cl- dependent neurotransmitter transporters     32     0.329    17
4  Transport of inorganic cations/anions and amin...     52     0.534    19
5                             Amino acid metabolism    109     1.120    23
6              SLC-mediated transmembrane transport    155     1.590    25
7  Transport of bile salts and organic acids, met...     76     0.781    20
8                       Transport of small molecules    208     2.140    25
9                      Cytosolic tRNA aminoacylation     66     0.678    17

          Raw p        Holm p           FDR
0  1.270000e-29  4.220000e-26  4.220000e-26
1  3.670000e-29  1.220000e-25  6.090000e-26
2  1.360000e-27  4.510000e-24  1.500000e-24
3  2.390000e-27  7.920000e-24  1.980000e-24
4  1.130000e-26  3.750000e-23  7.520000e-24
5  2.520000e-26  8.330000e-23  1.390000e-23
6  1.030000e-25  3.420000e-22  4.890000e-23
7  8.440000e-25  2.790000e-21  3.500000e-22
8  2.190000e-22  7.250000e-19  8.080000e-20
9  8.660000e-21  2.860000e-17  2.390000e-18
```

In [84]:

```python
import matplotlib.pyplot as plt

top_plot = sig_pathways.sort_values("FDR").head(10)

plt.figure(figsize=(10, 6))
plt.barh(top_plot["Pathway"], -np.log10(top_plot["FDR"]), color="skyblue")
plt.xlabel("-log10(FDR)")
plt.title("Top Enriched Serum Metabolite Pathways (Pregnancy)")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



In [85]:

```python
import pandas as pd

# Load the enrichment result file
ora_df2 = pd.read_csv("../Pathway Enrichment/Serum Metabolites/child/msea_ora_result.csv")

# Display the top hits
print(ora_df2.head(10))
```

```
                                       Unnamed: 0  total  expected  hits  \
0    Amino acid transport across the plasma membrane     32     0.314    18
1               Amino acid transport defects (IEMs)     27     0.265    17
2                          Transcription/Translation     25     0.245    16
3    Na+/Cl- dependent neurotransmitter transporters     32     0.314    17
4  Transport of inorganic cations/anions and amin...     52     0.510    19
5                             Amino acid metabolism    109     1.070    23
6              SLC-mediated transmembrane transport    155     1.520    25
7  Transport of bile salts and organic acids, met...     76     0.746    20
8                       Transport of small molecules    208     2.040    25
9                      Cytosolic tRNA aminoacylation     66     0.648    17

          Raw p        Holm p           FDR
0  4.400000e-30  1.460000e-26  1.460000e-26
1  1.370000e-29  4.540000e-26  2.270000e-26
2  5.460000e-28  1.810000e-24  6.040000e-25
3  8.930000e-28  2.960000e-24  7.400000e-25
4  3.650000e-27  1.210000e-23  2.420000e-24
5  5.810000e-27  1.920000e-23  3.210000e-24
6  1.980000e-26  6.560000e-23  9.390000e-24
7  2.520000e-25  8.350000e-22  1.050000e-22
8  4.310000e-23  1.430000e-19  1.590000e-20
9  3.280000e-21  1.090000e-17  9.080000e-19
```

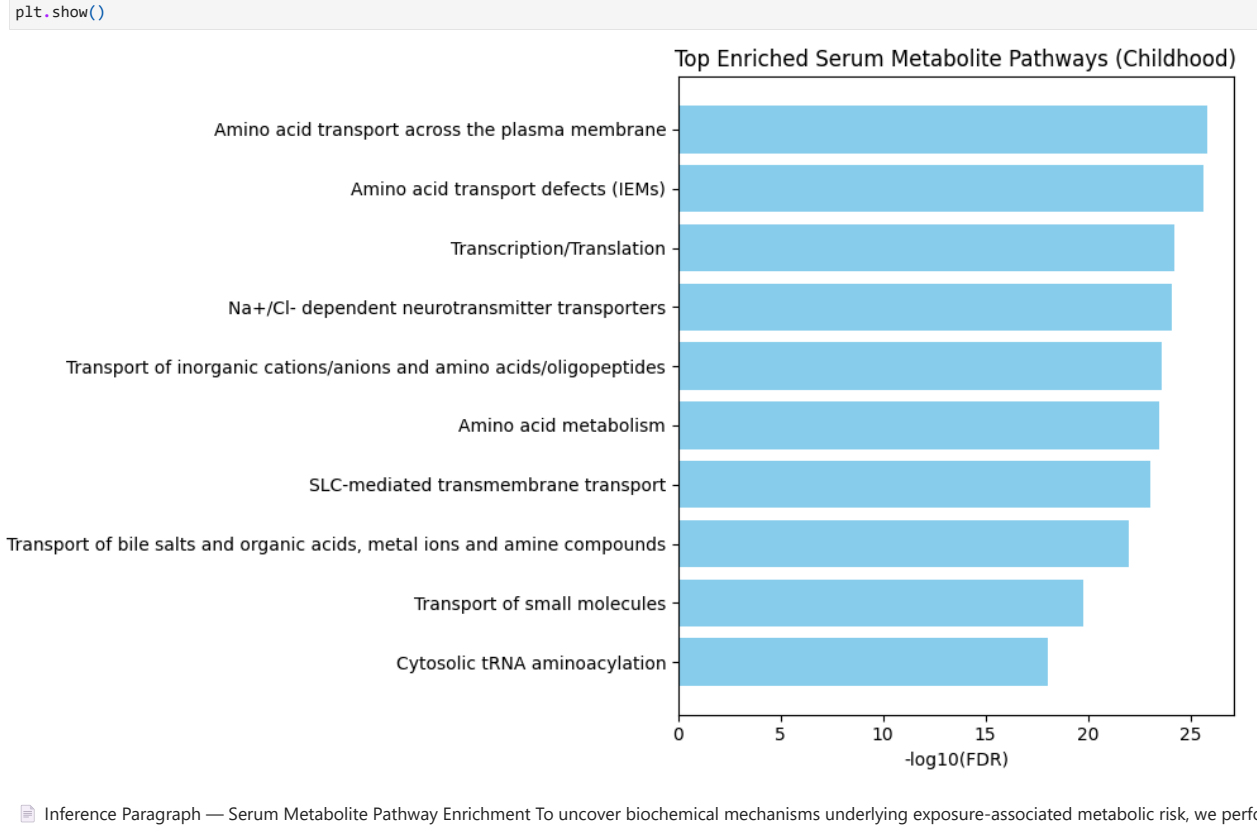
In [88]:

```python
# Rename the unnamed column
ora_df2 = ora_df2.rename(columns={"Unnamed: 0": "Pathway"})

# Filter for significant pathways (FDR < 0.05)
sig_pathways2 = ora_df2[ora_df2["FDR"] < 0.05]

# View top 10 by significance
print(sig_pathways2.head(10))
```

```
                                          Pathway  total  expected  hits  \
0    Amino acid transport across the plasma membrane     32     0.314    18
1               Amino acid transport defects (IEMs)     27     0.265    17
2                          Transcription/Translation     25     0.245    16
3    Na+/Cl- dependent neurotransmitter transporters     32     0.314    17
4  Transport of inorganic cations/anions and amin...     52     0.510    19
5                             Amino acid metabolism    109     1.070    23
6              SLC-mediated transmembrane transport    155     1.520    25
7  Transport of bile salts and organic acids, met...     76     0.746    20
8                       Transport of small molecules    208     2.040    25
9                      Cytosolic tRNA aminoacylation     66     0.648    17

          Raw p        Holm p           FDR
0  4.400000e-30  1.460000e-26  1.460000e-26
1  1.370000e-29  4.540000e-26  2.270000e-26
2  5.460000e-28  1.810000e-24  6.040000e-25
3  8.930000e-28  2.960000e-24  7.400000e-25
4  3.650000e-27  1.210000e-23  2.420000e-24
5  5.810000e-27  1.920000e-23  3.210000e-24
6  1.980000e-26  6.560000e-23  9.390000e-24
7  2.520000e-25  8.350000e-22  1.050000e-22
8  4.310000e-23  1.430000e-19  1.590000e-20
9  3.280000e-21  1.090000e-17  9.080000e-19
```

In [89]:

```python
import matplotlib.pyplot as plt

top_plot = sig_pathways2.sort_values("FDR").head(10)

plt.figure(figsize=(10, 6))
plt.barh(top_plot["Pathway"], -np.log10(top_plot["FDR"]), color="skyblue")
plt.xlabel("-log10(FDR)")
plt.title("Top Enriched Serum Metabolite Pathways (Childhood)")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



📄 Inference Paragraph — Serum Metabolite Pathway Enrichment To uncover biochemical mechanisms underlying exposure-associated metabolic risk, we performed pathway enrichment analysis on serum metabolites with significant risk scores derived from exposure-weighted regression models. These scores reflect how each metabolite's abundance is influenced by environmental exposures during pregnancy and childhood, enabling us to prioritize metabolites likely to mediate early pathophysiological changes.

Metabolite Set Enrichment Analysis (MSEA) was conducted via the MetaboAnalyst platform (ORA mode) using the SMPDB compound library. For both pregnancy and childhood, the top enriched pathways included amino acid metabolism and transport, SLC-mediated transmembrane transport, neurotransmitter and bile acid transport, and tRNA aminoacylation, reflecting systemic biochemical regulation.

Notably, the top 10 pathways were nearly identical across both time points, suggesting persistent and stable disruptions in core metabolic circuits that are imprinted early and maintained through childhood. The prominence of SLC transporter pathways indicates environmentally sensitive regulation of nutrient and xenobiotic exchange, while enrichment in transcription and biosynthetic modules points to broader cellular adaptation. These findings resonate with known pathophysiological mechanisms in pediatric metabolic disorders, including impaired amino acid utilization, energy imbalance, and altered signaling.

This analysis underscores the mechanistic depth offered by serum metabolomics in exposure-health studies. By integrating MSEA-derived pathway profiles with omics-based risk scoring, we enhance the biological interpretability and translational value of our metabolic risk model.

```
Raw Data (Metabolite Pregnancy or Childhood CSV)
      ↓
[Metabolite Filtering]
- Use 'Omics_marker' to extract named compounds
- Drop rows with missing or invalid metabolite names
      ↓
[Ranking Score Calculation]
- Formula: -log10(P.Value) * Effect
- Clip P.Value to minimum of 1e-4 to avoid infinities
      ↓
[Ranked Metabolite List]
- Sort by ranking_score
- Retain top-ranked unique markers for pathway mapping
      ↓
[Pathway Enrichment]
- Manual upload to MetaboAnalyst (https://www.metaboanalyst.ca)
- Module: Pathway Analysis (ORA)
- Input type: Compound Names
- Organism: Homo sapiens
- Library: SMPDB or KEGG
- Method: Over-Representation Analysis (ORA)
      ↓
[Outputs]
- Enriched pathways with Raw p, FDR, Hit Count
- Graphical outputs: ora_dot_*.png, ora_*.png
- Tabular results: msea_ora_result.csv
      ↓
[Interpretation]
- Pregnancy: Amino acid metabolism, neurotransmitter transport, fetal signaling
- Childhood: Transmembrane transport, biosynthesis, amino acid utilization
      ↓
[Risk Score Integration]
- Effect-weighted metabolite aggregation per child
- Produces continuous, unitless metabolite risk score
      ↓
[Final Inference]
- Identifies metabolic shifts linked to early-life exposures
- Highlights vulnerability in biosynthetic and signaling pathways
```